Phase 1: Project Planning & Management

Project: BloodLink | رابط الحياة
Program: DEPI (Digital Egypt Pioneers Initiative), 2026
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Deadline: 16 Oct 2026

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1. Project Proposal

Overview

BloodLink is a community website that connects people in Egypt who need blood with suitable donors nearby. Every request is backed by an official hospital paper. Donor and patient always meet at the hospital, and no money is involved.

Problem

- Blood requests are spread through Facebook and WhatsApp posts that reach people by chance.
- Nobody can tell if a post is real or still needed.
- Most people who see a post have the wrong blood type or live too far away.
- Contact details are posted publicly, which invites scams and requests for money.

Solution

Step	What happens
Donor joins	Uploads an official hospital certificate showing their blood type
Patient asks	Uploads an official hospital paper proving the patient needs blood
Paper check	Partner hospital paper: confirmed by the hospital, badge **Verified**. Other official hospital: checked by AI, badge **Paper looks valid**.
Matching	Compatible, available donors closest to the hospital are notified first, and the area grows in waves
Accept	The donor accepts. Phone numbers are shared only then.

Step	What happens
Meet	At the hospital. The hospital decides medically. Partner hospitals scan the donor's QR code to record the donation.

Objectives

1. Send each request only to donors who can actually help: compatible blood type, available, and nearby.
2. Show how much each request and donor can be trusted, using hospital papers and clear badges.
3. Protect users: phone numbers stay hidden until a donor accepts, meetings happen only at hospitals, and money requests are banned.
4. Deliver a working, deployed Arabic website that works well on phones and computers, by 30 Nov 2026.

Scope

In scope (core)	Out of scope
Accounts with 4 roles: donor, requester, partner hospital, admin	Examining donors or making any medical decision
Certificate and hospital paper upload	Guaranteeing a paper is genuine
AI paper check and partner-hospital confirmation	Payments of any kind
Matching by blood type, distance and trust level	SMS or WhatsApp Business API
Web push notifications in waves	Mobile apps (a possible future improvement; this project is a website only)
Accept or decline, with phone numbers hidden until acceptance	
QR check-in at partner hospitals; self-confirmation elsewhere	
Live request progress	
Admin: paper review, reports, bans, key numbers	
Arabic RTL interface that works on phones	

Planned extras (built only after checkpoint 2): an English version, a map view, a WhatsApp share card, donation history and charts.

Main deliverables

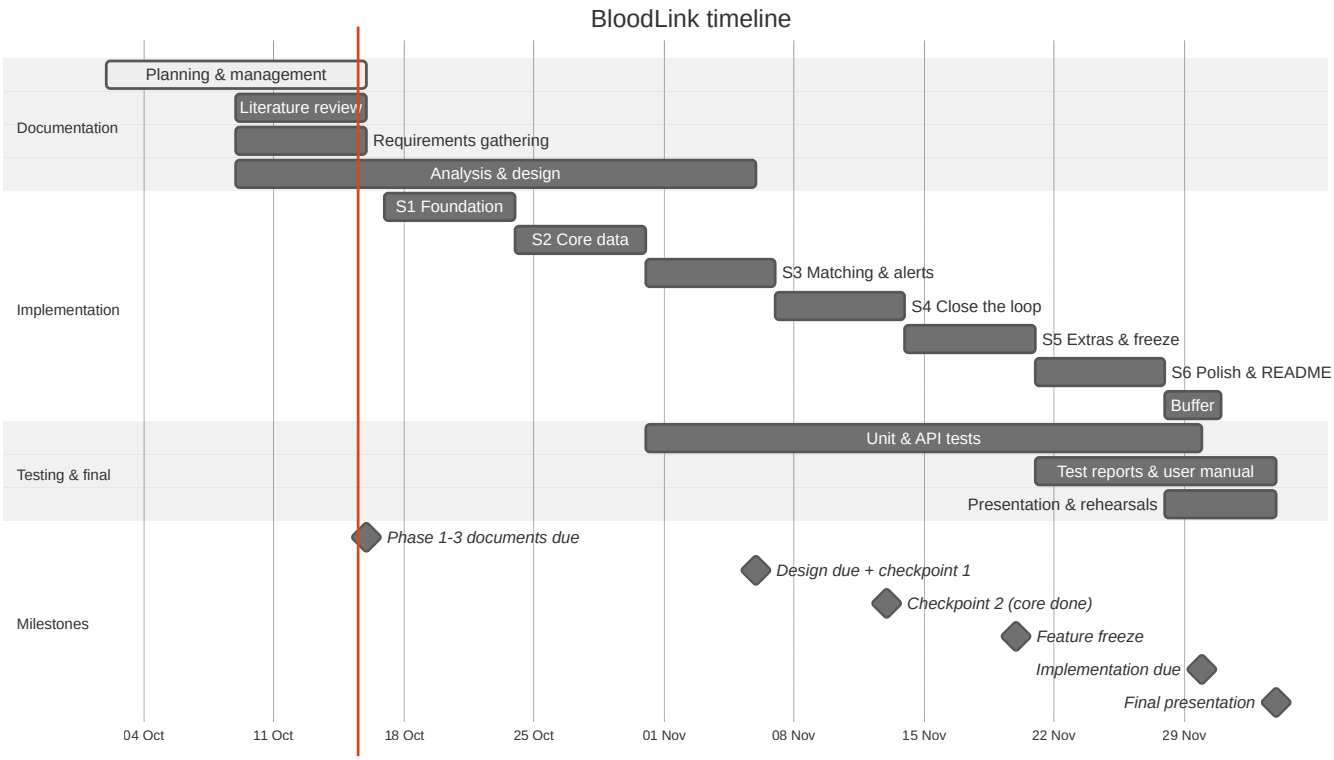
- Deployed website with demo data
- Source code on GitHub, with a README and API documentation
- Phase documents (planning, requirements, design, testing)
- Presentation, demo video and project infographic

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2. Project Plan

Timeline (Gantt chart)

The plan follows the DEPI deadlines. Implementation runs in six one-week sprints, Saturday to Friday, with a three-day buffer before the 30 Nov deadline.



Sprints

Sprint	Dates	Goal	Done when
S1 Foundation	17 to 23 Oct	Repo, TypeScript, Docker, deploy, data models, login with roles, Arabic UI kit	The website is deployed and a user can log in with a role
S2 Core data	24 to 30 Oct	Donor signup and certificate upload, request form and paper upload, partner confirmation	A request becomes active after the hospital confirms it
S3 Matching & alerts	31 Oct to 6 Nov	Matching with tests, AI paper check, push notifications, accept or decline, live progress	**Checkpoint 1:** the whole flow works roughly
S4 Close the loop	7 to 13 Nov	QR check-in at partner hospitals, self-confirmation elsewhere, waves, admin tools	**Checkpoint 2:** all core features are done
S5 Extras & freeze	14 to 20 Nov	Planned extras (if checkpoint 2 passed), security, bug fixes	**Feature freeze**
S6 Polish	21 to 27 Nov	Bug fixes only, README, API docs, demo data	Ready to submit

Milestones and deliverables

Date	Milestone	Deliverable
16 Oct	Phases 1 to 3 submitted	Planning, literature review and requirements documents
23 Oct	Skeleton live	Deployed website with login
6 Nov	Design submitted, checkpoint 1	Diagrams, database design, wireframes, the end-to-end flow working roughly
13 Nov	Checkpoint 2	All core features working
20 Nov	Feature freeze	No new features after this date
30 Nov	Implementation submitted	Source code, README, deployed link
4 Dec	Final presentation	Test reports, user manual, slides, demo video, infographic

Resource allocation

Time (worst case). Every member works about 10 hours a week for 8 weeks, from 9 Oct to 4 Dec.

	Hours
Capacity: 6 members × 10 h × 8 weeks	480
Weekly meetings (1 h each per week)	−48
One member loses a full week	−10
Available	**422**
Core development (estimates × 1.5 safety factor)	167
Planned extras (× 1.5)	59
Code reviews and integration	40
Documentation for all phases (× 1.5)	60
Research, presentation, video, rehearsals (× 1.5)	30
Planned	**356**
Spare	**≈ 66**

The spare hours cover delays and bugs. The extras start only after checkpoint 2, so the core is always finished first.

Role	Development & reviews	Docs & presentation	Total (of 72 h)
R1 Team Leader & DevOps (Ibrahim)	≈ 46 h	≈ 12 h	≈ 58 h
R2 Backend: Matching (Khaled)	≈ 50 h	≈ 8 h	≈ 58 h
R3 Backend: Papers (Mohammed)	≈ 50 h	≈ 8 h	≈ 58 h
R4 Frontend: Donor (Hager)	≈ 48 h	≈ 10 h	≈ 58 h
R5 Frontend: Requester (Zeyad)	≈ 48 h	≈ 10 h	≈ 58 h
R6 UI/UX & QA (Marawan)	≈ 24 h	≈ 42 h	≈ 66 h

Tools and services (cost: $0).

Need	Tool	Plan
Code and tasks	GitHub, GitHub Projects, GitHub Actions	Free
Frontend hosting	Vercel	Free (Hobby)
Backend hosting	Render (Docker)	Free, kept awake with an uptime ping
Database	MongoDB Atlas	Free (M0)
File storage	Cloudinary	Free tier
AI paper check	Mock mode for development, free AI tier for the demo with sample papers only	Free
Design	Figma	Free

At launch, the only real cost would be a paid AI model for real papers (roughly $3 to $12 per 1,000 papers).

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3. Task Assignment & Roles

Roles

The team has six roles, and each member takes one. The assignment below is **temporary** and may change once the team confirms it.

#	Role	Member	Main responsibilities	Best fit for someone who…
R1	**Team Leader & DevOps**	Ibrahim Hassan Anwar	Architecture, repo setup, CI, Docker, deployment, authentication and roles, data models, demo data, reviewing pull requests, final say on scope	is the strongest all-round developer and is comfortable with Git and Docker
R2	**Backend: Matching & Alerts**	Khaled Atef Salah	Matching engine and its unit tests, notification waves, push notifications, live updates (Socket.io)	likes logic, algorithms and Node.js
R3	**Backend: Papers & Trust**	Mohammed Eslam Mohammed	AI paper check (`checkPaper()`) and its prompt, trust badges, request states, QR check-in API, admin API (paper review, reports, bans)	likes APIs, databases and working with AI
R4	**Frontend: Donor**	Hager Waleed Shaker	Donor signup and certificate upload, notifications on the phone, accept or decline, QR screen, donation history	likes React and building screens for phones
R5	**Frontend: Requester & Hospital**	Zeyad Mohammed Mohammed	Request form and paper upload, live progress page, partner-hospital portal (confirm papers, scan QR), admin dashboard	likes React, forms and dashboards
R6	**UI/UX & QA**	Marawan Hesham Tolba	Wireframes, UI kit and Arabic RTL guidelines, test plan and bug reports, slides and infographic, lead presenter	has an eye for design, writes clearly and presents well

Document owners

Each phase has one owner who collects the work and submits it. Everyone contributes.

Phase	Owner	Contributors
1. Planning & management	R1	All
2. Literature review (feedback, improvements)	R6	R4
3. Requirements gathering	R6	R4, R5
4. Analysis & design	R2	R1: architecture, deployment, components R3: ERD, schema, class and state diagrams R4: DFD, activity diagram R5: sequence diagrams R6: wireframes, UI guidelines
5. Implementation (code, README)	R1	All
6. Testing & QA	R6	R2, R3: automated tests R4, R5: UI tests
7. Final presentation & reports	R6	R1: technical docs R4: demo video R5: user manual

How we work

- **Board:** GitHub Projects, with the columns To do, In progress, Review and Done.
- **Meetings:** a 1-hour planning meeting at the start of each sprint, and a short written update every two days.
- **Code:** one branch per feature, a pull request reviewed by one teammate, and CI must pass before merging.
- **Decisions:** after sprint 2, only the team leader can add new features to the plan.

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4. Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
1	A team member is unavailable for a week or more	Medium	High	Hours planned for the worst case. Two members on each side (backend and frontend) can cover for each other. The API is documented.
2	The work runs out of time	Medium	High	A fixed core scope, checkpoints on 6 and 13 Nov, a feature freeze on 20 Nov and a 3-day buffer.
3	Six people working on the same code cause conflicts or duplicated work	Medium	Medium	Each member owns clear folders and features, pull requests stay small, and there's a short written update every two days.
4	Documentation takes time away from coding	High	Medium	Each phase has one owner, diagrams are written as code in the repo, and documents are written while the feature is being built.

#	Risk	Likelihood	Impact	Mitigation
5	The AI marks a fake paper "looks valid"	Medium	Medium	The badge never claims a paper is genuine. Users accept terms saying they are responsible for their papers. Users can report, and abusers are banned. Partner hospitals give the stronger badge.
6	The free AI service is slow or down during the demo	Medium	High	Mock mode (`AI_PROVIDER=mock`), papers checked in advance and a recorded backup video.
7	Push notifications are hard to get working	Medium	Medium	Started early in sprint 3. Live updates on the website are the fallback. Demo on Android Chrome.
8	The free server sleeps and misses notification waves	High	Low	An uptime ping every 5 to 10 minutes.
9	No real hospital agrees to partner before the demo	High	Low	Demo hospitals have clearly fictional names. We never claim a partnership that doesn't exist.
10	Users' personal data or papers leak	Low	High	Papers are private files, phone numbers are hidden until a donor accepts, no secrets go in the repo, and only sample papers are sent to the free AI. We follow Egypt's Personal Data Protection Law (151/2020).
11	People ask for money or misuse the platform	Medium	High	Clear "no money" rules on every screen, a report button and admin bans.
12	Users see us as responsible for medical outcomes	Low	High	Terms of use say we only connect people. The hospital makes every medical decision.

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5. KPIs (Key Performance Indicators)

These are targets we set for the demo and a small pilot. They are not measured results yet.

Product

KPI	Target	How we measure
Time from request to first donor accepting	Under 30 minutes in the demo	Timestamps stored on each request
Requests that reach at least one accepting donor	≥ 70 % in the pilot	Admin dashboard
Notified donors who accept or decline	≥ 50 %	Notification and response logs
Requests that have a hospital paper	100 %	Enforced by the system
Abuse reports handled by an admin	100 % within 24 hours	Report status and timestamps

Technical

KPI	Target	How we measure
API response time	95 % of requests under 500 ms	Server logs
Uptime during demo week	≥ 99 %	Uptime monitor
Page load on a phone over 4G	Under 3 seconds	Lighthouse
Accessibility score	≥ 90	Lighthouse

KPI	Target	How we measure
Test coverage of core logic (matching, trust badges, request states)	≥ 70 %	Jest coverage report

Adoption (pilot)

KPI	Target	How we measure
Registered donors	50	Database count
Donors with an uploaded certificate	≥ 60 % of registered donors	Database count
Partner hospitals	At least 1 real partner talked to	Outreach log

Project delivery

KPI	Target
Phase documents submitted on time	7 of 7
Core features done by checkpoint 2 (13 Nov)	100 %
Pull requests reviewed before merging	100 %
Critical bugs open at submission	0